📚 **ENGLISH CODING**

BC001 · Python Foundations

Python Bootcamp · 100 Days — Days 1–16. Variables → OOP intro.

For parents: these are the key English coding words your child meets in this course, week by week. You don't need to code — just read a word together and ask your child to explain it back in their own words. The tiny example shows what it looks like in real code.

W1

WORD	WHAT IT MEANS	TINY EXAMPLE
`print()`	show text on the screen	`print("Hello")`
`string`	text wrapped in quotes	`"Band Name"`
`input()`	read what the user types	`name = input("Name? ")`

W2

WORD	WHAT IT MEANS	TINY EXAMPLE
`int() / float()`	turn text into a whole / decimal number	`int("12")   # 12`
`round()`	round a number to fewer decimals	`round(3.14159, 2)`
`f-string`	a string with variables inside { }	`f"Total: {bill}"`

W3

WORD	WHAT IT MEANS	TINY EXAMPLE
`if / elif / else`	choose a path based on a condition	`if age >= 18: ...`
`comparison operators`	compare two values (>, <, ==)	`score > 50`
`logical and / or`	combine two conditions	`if a and b:`

W4

WORD	WHAT IT MEANS	TINY EXAMPLE
`list`	an ordered group of values in [ ]	`moves = ["rock", "paper"]`
`random module`	Python tool for random numbers/choices	`import random`
`random.randint`	a random whole number between two ends	`random.randint(0, 2)`

W5

WORD	WHAT IT MEANS	TINY EXAMPLE
`for loop`	repeat once per item or count	`for i in range(8):`
`range()`	a sequence of numbers	`range(5)   # 0..4`
`.append()`	add an item to the end of a list	`chars.append(letter)`

W6

WORD	WHAT IT MEANS	TINY EXAMPLE
`function (def)`	a named, reusable block of code	`def greet():` `    print('Hi')`
`argument`	the value you pass into a function	`greet('Sam')`
`indentation`	spaces that show what's inside a block	`def f():` `    return 1`

W7

WORD	WHAT IT MEANS	TINY EXAMPLE
`while loop`	repeat as long as a condition is true	`while not won:` `    guess()`
`string slicing`	take part of a string by position	`word[0:3]`

W8

WORD	WHAT IT MEANS	TINY EXAMPLE
`modulo (%)`	the remainder after division	`7 % 3   # 1`
`.index()`	find the position of an item	`alpha.index("c")`

W9

WORD	WHAT IT MEANS	TINY EXAMPLE
`dictionary`	a key→value mapping in { }	`{"bid": 100}`
`key / value`	label and the data stored under it	`bids["Sam"] = 50`

W10

WORD	WHAT IT MEANS	TINY EXAMPLE
`nested dictionary`	a dictionary inside a dictionary	`{"latte": {"water": 200}}`
`type casting`	convert one data type to another	`float(amount)`

W11

WORD	WHAT IT MEANS	TINY EXAMPLE
`return value`	the result a function hands back	`return total`
`sum()`	add up all numbers in a list	`sum([5, 3, 2])`

W12

WORD	WHAT IT MEANS	TINY EXAMPLE
`scope`	where a variable can be seen/used	`local vs global`
`global`	use/modify a top-level variable inside a function	`global score`

W13

WORD	WHAT IT MEANS	TINY EXAMPLE
`traceback`	the error report Python prints	`IndexError: list index...`
`debugging`	finding and fixing mistakes	`print(x) to inspect`

W14

WORD	WHAT IT MEANS	TINY EXAMPLE
`game loop`	the while-true that runs the game	`while playing:`

W15

WORD	WHAT IT MEANS	TINY EXAMPLE
`data structure`	an organized way to hold data	`menu = {...}`

W16

WORD	WHAT IT MEANS	TINY EXAMPLE
`class`	a blueprint for making objects	`class Quiz:`
`object`	one thing built from a class	`q = Quiz()`
`__init__`	the setup method that runs on creation	`def __init__(self):`